Maya Chen

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Education

Northbridge University (Fictional) *Sep. 2022 - Jun. 2026*
BSc in Computer Science Boston, MA
GPA: 3.91/4.00, Class Ranking: Top 5%

Relevant Coursework: Machine Learning, Algorithms, Probability, Linear Algebra

Research Experience

Undergraduate Research Assistant - Robust Learning *Sep. 2024 - Jun. 2026*
Northbridge Reliable AI Lab | Advisor: Dr. Elena Brooks

- Implemented image-classification baselines in PyTorch and evaluated robustness under distribution shifts.
- Compared augmentation and calibration methods across 12 corruption settings using five random seeds.
- Reduced expected calibration error from 8.4% to 5.1% on the held-out evaluation split.
- Defined corruption severity groups and kept preprocessing, checkpoints and inference settings fixed across comparisons.
- Conducted ablations on label smoothing, augmentation strength and temperature scaling to separate their effects.
- Reported clean-data accuracy alongside robustness metrics to make accuracy-calibration tradeoffs explicit.
- Wrote data-loading tests and saved configurations, environment versions and seed-level results for replication.

Summer Research Student - Dataset Quality *Jun. 2024 - Aug. 2024*
Northbridge Data Science Center | Advisor: Dr. Daniel Reed

- Audited 18,000 labels and documented annotation disagreements in an internal research report.
- Created stratified data splits and leakage checks for reproducible model comparison.
- Designed an annotation review protocol covering ambiguous labels, near-duplicates and class imbalance.
- Reviewed disagreements with two student annotators and recorded decisions without overwriting the original labels.
- Compared class distributions before and after filtering and identified categories with insufficient coverage.

Professional Experience

Cedar Analytics (Fictional) *Jun. 2025 - Aug. 2025*
Machine Learning Intern Remote

- Built feature-validation checks and experiment summaries for an internal demand-forecasting prototype.
- Implemented time-based validation and compared seasonal baselines before testing tree-based forecasting models.
- Investigated missing values, delayed records and feature drift, documenting risks for deployment.
- Prepared a handover notebook explaining evaluation assumptions and limitations to the engineering team.

Selected Projects

Experiment Reproducibility Toolkit *2025*
Python, PyTorch, YAML, Git

- Built configuration-based runners, seed tracking and aggregated confidence-interval reports.
- Documented benchmark setup and added tests for split integrity and missing metrics.
- Added resumable execution and explicit failure records to avoid silently excluding unsuccessful experiments.
- Created a tutorial with a small synthetic dataset and a worked example of interpreting uncertainty.

- Separated dataset configuration from training code and checked repeated runs for consistent split membership.

Calibration Diagnostics Dashboard *Jan. 2026 - May 2026*

Python, Streamlit, pandas, Plotly

- Built reliability diagrams and class-level error summaries from saved prediction files.
- Added subgroup filters and displayed sample sizes to discourage conclusions from sparse categories.
- Compared expected calibration error under multiple binning schemes and documented estimator sensitivity.
- Wrote tests for malformed probability vectors, missing labels and classes absent from the evaluation split.

Small-Data Learning Study *2024*

Python, PyTorch, scikit-learn

- Designed a course study comparing linear models, shallow networks and pretrained representations under restricted label budgets.
- Evaluated five training-set sizes with fixed class balance and repeated subsampling.
- Compared validation-based model selection with a deliberately overfit baseline to demonstrate evaluation leakage.
- Recorded compute time and parameter count alongside classification accuracy.
- Wrote a project report explaining why gains on one dataset did not establish general superiority.

Model Error Review Workspace *2025*

Python, SQLite, pandas

- Built a local workspace for reviewing misclassified examples and grouping errors by category.
- Designed an annotation table retaining model version, confidence, review decision and reviewer notes.
- Added export checks to prevent duplicate examples from entering both training and evaluation collections.
- Analyzed error clusters with the research team and proposed targeted data-review tasks.
- Documented that qualitative error categories were exploratory and required independent validation.

Technical Skills

Programming Languages: Python, C++, SQL

Frameworks & Libraries: PyTorch, NumPy, pandas, scikit-learn

Systems & Tools: Linux, Git, Docker

Specializations: Robust learning, experiment design, model calibration

Honors & Awards

Northbridge Undergraduate Research Scholarship 2025 – *Supported a year-long independent research project.*

Faculty Academic Excellence Award 2024 – *Recognized performance in mathematics and computing.*

Research Poster Presentation Commendation 2026 – *Recognized a clear presentation of methods and experimental limitations.*